

allowing users to access data on Google Storage.

- **Unity Machine Learning Agents:** The Unity ML Agents SDK allows developers and researchers to transform games and simulations developed via Unity's editor into environments, where intelligent agents can be trained effectively using deep reinforcement learning.
- **Apache PredictionIO:** This is an open source ML stack integrated with Apache Spark, MLlib, HBase, Spray and Elastic Search to create predictive engines for all sorts of ML tasks.
- **Amazon Machine Learning:** The Amazon ML recommendation engine is used to power Amazon Echo/Dot, powered by Alexa, Drone-Prime Air, Amazon Go and AWS based cloud services.
- **Microsoft Distributed Machine Learning Toolkit:** This toolkit by Microsoft provides a framework for training models of ML on Big Data. The toolkit contains both algorithm and system innovations to make tasks on Big Data highly scalable, efficient and flexible.

2. The R programming language

R is a free and open source ML language supporting statistical computing and the graphics language on a wide range of operating systems. It provides diverse statistical functionalities like linear and non-linear modelling, classical statistical tests, time-series analysis, classification, clustering and advanced graphical techniques. It is a highly dynamic, scalable and extensible language that provides advanced features for data manipulation, calculation, graphics display, array calculation, data analysis tools, etc. It is a programming language that also contains conditions, loops and other capabilities.

R contains more than 11,000 packages of various kinds, which are available through the Comprehensive R Archive Network (CRAN), Bioconductor, Omegahat, GitHub and other repositories.

3. Emerging trends in blockchains and Bitcoin

Blockchain technology is rapidly undergoing intense development due to great interest from academia and the industry sector. A blockchain is regarded as a shared, open source transactional database for tracking transactions of digital currency like Bitcoin.

According to Don and Alex Tapscott, "The blockchain is an incorruptible digital ledger of economic transactions that can be programmed to record not just financial transactions but virtually everything of value."

A blockchain is a continuously growing list of records called blocks which are linked and secured using cryptography, with every block having a hash pointer as a link to the previous block, a time stamp and transaction data.

Blockchain platforms

- **ERIS:** This allows everyone to create their own secure, low-cost application (that can run anywhere) using a

blockchain and smart contract technology.

- **HydraChain:** This is an extension to the Ethereum platform, with added support for creating permissioned distributed ledgers.
- **MultiChain:** This enables organisations to rapidly design, deploy and operate distributed ledgers.
- **OpenChain:** This is open source distributed ledger technology suited for organisations working on issues and managing digital assets in a highly secure and flexible manner.
- **Ethereum Project:** This is a decentralised platform running smart contracts: Applications that run as such are programmed without any limitations of downtime, censorship, fraud or third party involvement.
- **Hyperledger:** This is an open source collaborative effort to advance cross-industry blockchain technology and ensure transparency, interoperability and support to bring blockchain technologies for commercial adoption.

4. Open source and the Internet of Things (IoT)

The Internet of Things (IoT) is highly fragmented and changing continuously. Open source is playing a crucial role in creating IoT platforms as well as ready-made prototypes in terms of development boards for R&D and automation.

IoT standards, together with Artificial Intelligence (AI), are controlling and interpreting a wide range of activities in a smart manner. Without the use of open source technologies, there is no Web. IoT requires the same level of ubiquitous common access in its core functions and to the Web for shared accessibility. IoT deployments in the near future are expected to connect and integrate globally with billions of devices, assets, sensors and end points.

IoT platforms are regarded as the middleware layer between IoT devices or end points and services that consume data outputs. These platforms offer sophisticated end point management to control the devices.

Open source IoT platforms

- **Kaa IoT:** This is an efficient, open source and cloud based IoT platform which enables data management for connected objects and back-end infrastructure by providing server and endpoint SDK components.
- **SiteWhere:** This provides ingestion, storage, processing and integration of device data. It runs on core servers provided by Apache Tomcat, and contains MongoDB and HBase implementations.
- **ThingSpeak:** This allows users to collect and store sensor data in the cloud and on popular IoT applications development platforms. It works well with Arduino, ESP8266, BeagleBone, Raspberry Pi, MATLAB, etc.
- **DeviceHive:** This provides Docker and Kubernetes deployment options. It has the power to connect to any device or hacker board via Rest APIs, WebSockets, MQTT, etc.